

The next-generation DNA vaccine platforms and delivery systems: advances, challenges and prospects.

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Vaccines have proven effective in the treatment and prevention of numerous diseases. However, traditional attenuated and inactivated vaccines suffer from certain drawbacks such as complex preparation, limited efficacy, potential risks and others. These limitations restrict their widespread use, especially in the face of an increasingly diverse range of diseases. With the ongoing advancements in genetic engineering vaccines, DNA vaccines have emerged as a highly promising approach in the treatment of both genetic diseases and acquired diseases. While several DNA vaccines have demonstrated substantial success in animal models of diseases, certain challenges need to be addressed before application in human subjects. The primary obstacle lies in the absence of an optimal delivery system, which significantly hampers the immunogenicity of DNA vaccines. We conduct a comprehensive analysis of the current status and limitations of DNA vaccines by focusing on both viral and non-viral DNA delivery systems, as they play crucial roles in the exploration of novel DNA vaccines. We provide an evaluation of their strengths and weaknesses based on our critical assessment. Additionally, the review summarizes the most recent advancements and breakthroughs in pre-clinical and clinical studies, highlighting the need for further clinical trials in this rapidly evolving field.

Ethics of DNA Vaccine Transfer for Clinical Research.

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Several experimental human DNA vaccines are currently undergoing Phase I, II, and III clinical trials in order to investigate their efficacy and safety. Human clinical trials must follow guidelines and procedures that have been approved by the regulatory authorities and ethics committees. Ethical clinical research is much more than applying an informed consent to participants. In this chapter we will review the ethical standards and provide a framework to evaluate and design ethical clinical research. Despite being universal standards supported by universal guidelines, they must be adapted to the conditions in each country where the clinical research is being conducted.

DNA vaccines and intradermal vaccination by DNA tattooing.

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Over the past two decades, DNA vaccination has been developed as a method for the induction of immune responses. However, in spite of high expectations based on their efficacy in preclinical models, immunogenicity of first generation DNA vaccines in clinical trials was shown to be poor, and no DNA vaccines have yet been licensed for human use. In recent years significant progress has been made in the development of second generation DNA vaccines and DNA vaccine delivery methods. Here we review the key characteristics of DNA vaccines as compared to other vaccine platforms, and recent insights into the prerequisites for induction of immune responses by DNA vaccines will be discussed. We illustrate the development of second generation DNA vaccines with the description of DNA tattooing as a novel DNA delivery method. This technique has shown great promise both in a small animal model and in non-human primates and is currently under clinical evaluation.

Live Bacterial Vectors-A Promising DNA Vaccine Delivery System.

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Vaccination is one of the most successful immunology applications that has considerably improved human health. The DNA vaccine is a new vaccine being developed since the early 1990s. Although the DNA vaccine is promising, no human DNA vaccine has been approved to date. The main problem facing DNA vaccine efficacy is the lack of a DNA vaccine delivery system. Several studies explored this limitation. One of the best DNA vaccine delivery systems uses a live bacterial vector as the carrier. The live bacterial vector induces a robust immune response due to its natural characteristics that are recognized by the immune system. Moreover, the route of administration used by the live bacterial vector is through the mucosal route that beneficially induces both mucosal and systemic immune responses. The mucosal route is not invasive, making the vaccine easy to administer, increasing the patient's acceptance. Lactic acid bacterium is one of the most promising bacteria used as a live bacterial vector. However, some other attenuated pathogenic bacteria, such as *Salmonella* spp. and *Shigella* spp., have been used as DNA vaccine carriers. Numerous studies showed that live bacterial vectors are a promising candidate to deliver DNA vaccines.

Lipid nanoparticle-encapsulated DNA vaccine robustly induce superior immune responses to the mRNA vaccine in Syrian hamsters.

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DNA vaccines for infectious diseases and cancer have been explored for years. To date, only one DNA vaccine (ZyCoV-D) has been authorized for emergency use in India. DNA vaccines are inexpensive and long-term thermostable, however, limited by the low efficiency of intracellular delivery. The recent success of mRNA/lipid nanoparticle (LNP) technology in the coronavirus disease 2019 (COVID-19) pandemic has opened a new application for nucleic acid-based vaccines. Here, we report that plasmid encoding a trimeric spike protein with LNP delivery (pTS/LNP), similar to those in Moderna's COVID-19 vaccine, induced more effective humoral responses than naked pTS or pTS delivered via electroporation. Compared with TSmRNA/LNP, pTS/LNP immunization induced a comparable level of neutralizing antibody titers and significant T helper 1-biased immunity in mice; it also prolonged the maintenance of higher antigen-specific IgG and neutralizing antibody titers in hamsters. Importantly, pTS/LNP immunization exhibits enhanced cross-neutralizing activity against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) variants and protects hamsters from the challenge of SARS-CoV-2 (Wuhan strain and the Omicron BA.1 variant). This study indicates that pDNA/LNPs as a promising platform could be a next-generation vaccine technology.

Dendrimer-Mediated Delivery of DNA and RNA Vaccines.

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DNA and RNA vaccines (nucleic acid-based vaccines) are a promising platform for vaccine development. The first mRNA vaccines (Moderna and Pfizer/BioNTech) were approved in 2020, and a DNA vaccine

(Zydus Cadila, India), in 2021. They display unique benefits in the current COVID-19 pandemic. Nucleic acid-based vaccines have a number of advantages, such as safety, efficacy, and low cost. They are potentially faster to develop, cheaper to produce, and easier to store and transport. A crucial step in the technology of DNA or RNA vaccines is choosing an efficient delivery method. Nucleic acid delivery using liposomes is the most popular approach today, but this method has certain disadvantages. Therefore, studies are actively underway to develop various alternative delivery methods, among which synthetic cationic polymers such as dendrimers are very attractive. Dendrimers are three-dimensional nanostructures with a high degree of molecular homogeneity, adjustable size, multivalence, high surface functionality, and high aqueous solubility. The biosafety of some dendrimers has been evaluated in several clinical trials presented in this review. Due to these important and attractive properties, dendrimers are already being used to deliver a number of drugs and are being explored as promising carriers for nucleic acid-based vaccines. This review summarizes the literature data on the development of dendrimer-based delivery systems for DNA and mRNA vaccines.

Polymeric Materials for Gene Delivery and DNA Vaccination.

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Gene delivery holds great potential for the treatment of many different diseases. Vaccination with DNA holds particular promise, and may provide a solution to many technical challenges that hinder traditional vaccine systems including rapid development and production and induction of robust cell-mediated immune responses. However, few candidate DNA vaccines have progressed past preclinical development and none have been approved for human use. This Review focuses on the recent progress and challenges facing materials design for nonviral DNA vaccine drug delivery systems. In particular, we highlight work on new polymeric materials and their effects on protective immune activation, gene delivery, and current efforts to optimize polymeric delivery systems for DNA vaccination.

Intramuscular DNA vaccination protocols mediated by electric fields.

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Vaccination is historically one of the most important methods for preventing infectious diseases in humans and animals. New insights in the biology of the immune system allow a more rational design of vaccines, and new vaccination strategies are emerging. DNA vaccines have been proposed as a promising approach for introducing foreign antigens into the host for inducing protective immunity against infectious and cancer diseases. Nevertheless, because of their poor immunogenicity, plasmid DNA vaccination strategies need further implementations. Recent data suggest electrotransfer as a useful tool to improve DNA-based vaccination protocols, being able to stimulate both the humoral and cellular immune responses. In preclinical trials, gene electrotransfer is successfully used in prime-boost combination protocols and its tolerability and safety has been demonstrated also in Phase I clinical trials. In this chapter, we report a short comment supporting electrotransfer as an effective strategy to improve DNA-based vaccination protocols and describe the vaccination procedures by plasmid DNA in combination with electrotransfer and hyaluronidase pretreatment in use in our laboratory.

Electroporation: a promising method for the nonviral delivery of DNA vaccines in humans?

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The without a doubt major obstacle for making DNA vaccines a commercial success is delivery. If delivery cannot be made simple, cheap and effective, DNA vaccines may not become a viable option for human use. Numerous clinical trials have confirmed that a standard needle and syringe simply do not do the job, i.e., delivering the DNA payload inside the cell. Having recognized this shortcoming, investigators have developed several new approaches for DNA vaccine delivery. In particular, new types of delivery devices, originally intended for in vitro use, have been applied for in vivo delivery. These include particle bombardment or biolistic delivery, and in vivo electroporation (EP). Importantly, both techniques seem to overcome the size barrier, meaning that they work in both mice and larger animals. In vivo EP has the key features of improved DNA uptake, increased antigen expression and a local inflammation. These factors are essential to make DNA vaccines effective in a larger host. Early data from clinical trials with DNA vaccines delivered by in vivo EP are cautiously promising. Thus, we may be entering a new era of DNA vaccination where we start to see clinical effects in humans; however, these may also be accompanied by side effects, as the vaccines become more effective.

Drug-induced acute interstitial nephritis: pathology, pathogenesis, and treatment.

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Drug-induced acute interstitial nephritis (DAIN) is a common cause of acute kidney injury and often presents as an unexplained rise in serum creatinine level. Kidney biopsy is therefore frequently required to make a definitive diagnosis. The hallmark pathologic features of DAIN are interstitial edema, interstitial inflammation, and tubulitis with a predominance of CD4+ T lymphocytes and mononuclear cells, with variable numbers of eosinophils. This is a result of a type B idiosyncratic non-immunoglobulin-E-mediated immune reaction marked by cell-mediated immune injury to the renal tubulointerstitium. The drug becomes immunogenic via various mechanisms such as haptenization, antigen mimicry, and neo-antigen formation. Renal interstitial dendritic cells, and renal tubular epithelial cells play an important role in further propagating this immunologic injury. Acute DAIN can progress within days to weeks to a chronic form triggered by fibroblast activation and manifested as interstitial fibrosis and tubular atrophy. The mainstay of treatment of DAIN is discontinuation of the offending drug. Incomplete renal recovery is seen in one-third of the patients and depends on the duration of injury prior to diagnosis. Use of steroids for treatment of DAIN makes biological sense, but lack of randomized controlled trials and conflicting data from retrospective studies makes the approach unclear. Positive effects include faster recovery of kidney function, more complete recovery with less chronic kidney disease, and reduced need for chronic dialysis. Therefore, it seems reasonable to employ corticosteroids in patients that do not rapidly improve 3 to 5 days following discontinuation of the offending agent.